Speech Recognition and Large Language Models for Monitoring Knowledge Acquisition from Teacher–Student Dialogue: A Literature Review

David Khamitskiy, Tairkhan Yermek, Yermakhan Khamitov

***Abstract*—This review examines the evidence for an intelligent system that estimates students' knowledge from recorded teacher–student lessons in English, Russian, and mixed English–Russian speech. The system cleans the audio, transcribes it, separates the speakers, uses a large language model (LLM) to rate the student's understanding of each topic, and prepares reports for the teacher, the student, and the parent. The review covers 19 peer-reviewed journal articles (12 research studies and 7 reviews) on speech recognition, diarization, LLM-based dialogue analysis and answer grading, knowledge tracing, and explainable AI, and 10 technical sources on the models and methods used in the system. Speech technologies are usable, but accuracy depends on the task, the speakers, and the language: the word error rate of a Whisper-based system was 9.65% for English and 11.86% for Russian against 26.32% for Kazakh, which is why Kazakh is outside the first version, and diarization error in noisy classroom work was about 34%. LLMs agree with human coders at the category level ($\kappa = 0.69$) but not on fine-grained codes, and grade short answers with an F1 of 0.61–0.74. Parameter-efficient fine-tuning makes it possible to adapt an open LLM on one GPU, but no reviewed study applies it to rating understanding. Explanations increase teachers' trust. Six research gaps are identified, and requirements for the system are derived from them.**

***Index Terms*—automatic speech recognition, speaker diarization, large language models, classroom dialogue, knowledge assessment, knowledge tracing, fine-tuning, explainable AI, code-switching.**

I. Introduction

WHAT a student says during a lesson is often the earliest evidence of whether a topic has been understood. Answers, hesitations, repeated questions, and corrections show misconceptions earlier than a test does. Teachers rarely have time to review recordings, so this evidence is usually lost. Advances in automatic speech recognition (ASR) and large language models (LLMs) make it possible to convert spoken lessons into structured data and to analyze them automatically [1]–[3].

The review informs the design of a system intended for online schools, tutoring centers, and independent tutors in Kazakhstan. A teacher uploads course materials and then the recording of a one-to-one lesson; group lessons are left for later. In the first version, the lessons are held in English, often with explanations in Russian, or in Russian; Kazakh is deferred for the reasons given in Section III. The system cleans the audio, transcribes it, separates teacher and student speech, and passes an anonymized transcript to an LLM. The LLM finds in the lesson the topics from the teacher's materials and rates the student's understanding of each topic with time-coded evidence from the dialogue. The ratings are combined, at a later stage, with the results of short quizzes into a per-topic index by fixed rules, and the indices form a knowledge map. Separate reports are produced for the teacher, the student, and the parent. The first version uses ready-made models, including a cloud LLM; in a second phase, an open LLM is fine-tuned on corrections made by teachers and compared with the cloud model.

Each stage of this pipeline can fail in ways that later stages inherit. A misrecognized word or an answer attributed to the wrong speaker changes what the LLM receives, and a report that teachers do not trust will not be used. The review is therefore organized around four questions:

- **RQ1:** How reliable are speech recognition and speaker diarization as a source of educational evidence, and how does reliability depend on language and recording conditions?
- **RQ2:** How well can LLMs analyze classroom dialogue and judge students' answers, and what is known about adapting an open LLM to such a task with little annotated data?
- **RQ3:** How can repeated evidence be turned into an estimate of a student's knowledge?
- **RQ4:** What does the literature say about trust, human oversight, and fairness in teacher-facing AI?

The answers also show which gaps remain for an end-to-end system of this kind.

Section II describes the review method. Sections III–VI answer RQ1–RQ4. Section VII compares the studies, identifies research gaps, and derives requirements for the system. Section VIII concludes.

II. Review Method

The review is a structured narrative review. It is purposive and not exhaustive: the aim was a compact evidence base covering every stage of the planned system. Searches were run in 2026 in Google Scholar and on the platforms of Frontiers, npj, Wiley, ScienceDirect, Springer, MDPI, Zenodo, and the Scientific Journal of Astana IT University, using terms on classroom dialogue analysis, LLMs in education, speech recognition and diarization in education, Kazakh ASR, short-answer grading, knowledge tracing, and explainable AI. Journal articles were included if they were (I1) peer-reviewed,

The authors are with Astana IT University, Astana, Kazakhstan.

(I2) connected to at least one stage of the system, (I3) empirical or methodological studies or reviews, and (I4) written in English. The core group contains 19 articles: 12 research studies (Table I) and 7 reviews (Table II). Eighteen were published in 2022–2026; one older review [4] was kept as a foundational map of the field.

A further search, run after the language scope of the system was set to English, Russian, and mixed speech and fine-tuning was added to the plan, covered arXiv, the ACL Anthology, and the ISCA Archive. It produced 10 technical sources, conference papers and preprints that describe models, datasets, and methods used or planned in the system [5]–[14]. Two of them [9], [10] are preprints.

Each study was appraised qualitatively for transparency of method, comparison with a human reference, treatment of fairness, and transferability to the setting of the system. Numerical results are quoted as reported by the original authors.

III. Speech Technologies as Evidence (RQ1)

A. Task, Speakers, and Fairness

Spoken interaction can serve as machine-readable evidence, but accuracy depends on the task, the speakers, and the language. Petscher et al. [15] compared a commercial speech-verification system with human raters on the responses of 429 kindergarten children. At the most liberal scoring threshold, mean item-level agreement was $\kappa = 0.15$ for phoneme blending, 0.52 for word reading, and 0.70 for expressive vocabulary. Agreement decreased with stricter thresholds and differed between Black and White students on some tasks. Speech verification checks a known target word, so the results transfer to open dialogue only indirectly, but they show that a single aggregate figure can hide large variation. Bailey et al. [16] frame the problem as one of validity: systems trained on data with few examples of varieties other than General American English score the speech of some children less accurately.

B. Languages of the System

The recognition component is based on Whisper [5], trained by its authors on 680,000 hours of multilingual data and transferable to standard benchmarks without dataset-specific fine-tuning. Direct evidence on the three languages comes from Makhataeva et al. [18], who evaluated Whisper large-v3-turbo fine-tuned by ISSAI. In their user study, 38 participants from one university read texts aloud in three languages. The system WER was 26.32% for Kazakh, 11.86% for Russian, and 9.65% for English, and the differences were significant. (The abstract of the article states the opposite; its Results section is followed here.) No participant reported English as a first language. Mussakhojayeva et al. [19] trained 22 ASR models for ten Turkic languages. For Kazakh, the best WER was 5.7% on the Kazakh Speech Corpus (KSC) test set and 28.6% on the Common Voice test set; the values come from different models, but they show how strongly the result depends on the test corpus.

Both studies use read or crowd-sourced speech and not lessons, and spontaneous lesson speech can be expected to produce higher error rates. The evidence on Kazakh, together with the absence of studies on Kazakh lessons, is the reason why Kazakh and Kazakh–Russian speech are outside the first version of the system.

C. Speaker Attribution

Wang et al. [17] optimized diarization for noisy spontaneous classroom group discussions. The best system reached a diarization error rate (DER) of about 34% and distinguished teacher from child speech with 69.17% accuracy. The diarization component of the proposed system is based on pyannote.audio [6], an open-source toolkit with pre-trained models for speech detection, speaker change, overlapped speech, and speaker embedding. One-to-one lessons contain two speakers and should be easier than group discussions, but no result for one-to-one English or Russian lessons was found.

D. Recording Conditions and Recognition Errors

Noise suppression does not always help recognition. Iwamoto et al. [7] identified the artifacts introduced by speech enhancement as the main cause of ASR degradation in noise, so the effect of denoising has to be measured by WER on the target recordings. Recognition errors also include invented text: Koenecke et al. [8] found that about 1% of Whisper transcriptions contained entire phrases absent from the audio, more often for speakers with a larger share of non-vocal duration. The study concerns speakers with aphasia, but a student who pauses before an answer produces similar audio, so pauses need separate handling.

E. Mixed English–Russian Speech

A systematic review of code-switching in end-to-end ASR [9] documents the languages, datasets, metrics, and models used. Zhao et al. [10] name language confusion caused by accents, similar sounds, and seamless switches as a challenge, and adapting Whisper on the SEAME corpus (Mandarin–English) reduced the mixed error rate by 4.1% and 7.2% relative to the baseline on two development sets. Both sources are preprints. None of the examined sources reports the accuracy of Whisper on English–Russian speech, so accuracy targets for mixed lessons are preliminary and have to be measured separately. Overall, an ASR transcript is an intermediate result whose errors propagate, and published WER figures say little about the accuracy on a new kind of recording.

IV. LLM Analysis of Dialogue and Adaptation of an Open Model (RQ2)

Two studies tested LLMs as coders of classroom dialogue. Long et al. [1] compared a customized GPT-4 model with expert coding of middle-school mathematics and Chinese lessons. They report significant time savings and high coding consistency, with minor discrepancies. Xu et al. [2] coded more than 400 Chinese talk turns from synchronous online mathematics lessons. At the category level, GPT-4o reached

TABLE I
RESEARCH STUDIES INCLUDED IN THE REVIEW (CORE GROUP)

#	Study	Data / context	Method	Key result (as reported)
1	Petscher et al. [15]	429 kindergarten students, USA; three word-level tasks	Commercial speech verification vs. human raters; Cohen's κ; Black/White subgroups	Mean item-level κ at the most liberal threshold: 0.15 (blending), 0.52 (word reading), 0.70 (vocabulary)
2	Bailey et al. [16]	Child language and literacy assessment, under-served U.S. students	Validity framing (Interpretation/Use Argument); experiments with ASR and NLP on child speech	Training data with few non-General American English varieties harms scoring validity; favorable results reported for African American English speakers
3	Wang et al. [17]	Noisy spontaneous classroom group discussions	ECAPA-TDNN embeddings + Whisper; design-choice comparison	DER $\approx$ 34%; r = 0.62 for student speaking time; teacher/child accuracy 69.17%
4	Makhataeva et al. [18]	38 participants (one university) reading Kazakh, Russian, and English texts	Whisper large-v3-turbo fine-tuned by ISSAI; WER/CER; ANOVA	Benchmark Kazakh WER 21.55% $\rightarrow$ 8.84%; system WER in the user study 26.32% (kk), 11.86% (ru), 9.65% (en)
5	Mussakhojayeva et al. [19]	Ten Turkic languages; KSC, Common Voice, and other corpora	13 monolingual and 9 multilingual ASR models	Best Kazakh WER 5.7% (KSC-based test set) vs. 28.6% (Common Voice test set); result depends on the test corpus
6	Long et al. [1]	Middle-school mathematics and Chinese lessons	Customized GPT-4 vs. expert coders	Significant time savings; high coding consistency; minor discrepancies
7	Xu et al. [2]	400+ Chinese talk turns, synchronous online mathematics	ChatGPT and GPT-4o vs. human standard	Category level: Cohen's κ = 0.69 (GPT-4o), 0.33 (ChatGPT); fine-grained codes unsatisfactory for both
8	Kortemeyer [20]	SciEntsBank, Beetle; 504 items, 14,186 answers	GPT-4 without task-specific training, with and without reference answers	2-way F1 0.744 and 0.611 (0.731 and 0.651 without reference); comparable to hand-engineered models, below pre-trained LLMs with specialized training
9	Sarsa et al. [21]	Public datasets and a new programming dataset	DKT, LSTM-DKT, DKVMN, SAKT vs. baselines including BKT	Deep models generally better, but margins often small; simple baselines can win on accuracy; results depend on hyperparameters, metrics, seeds, hardware
10	Cukurova et al. [22]	One-to-one online primary mathematics tutoring; 2,250 min, 44 tutors	CM-SPAM sequence mining; J48, JRIP classifiers	F-measure 0.89–0.98 for separating effective from less effective sessions
11	Aldino et al. [23]	21 higher-education teachers writing feedback	BERT classifier + ChatGPT; logs, interviews	BERT suggestions often adopted; ChatGPT drafts extensively revised; concerns about tone and trust
12	Wang et al. [24]	59 pre-service teachers (30 vs. 29)	AI dialogue analysis with vs. without explanations	Higher trust and acceptance with explanations; no significant difference in cognitive load

TABLE II
REVIEW PAPERS INCLUDED IN THE REVIEW (CORE GROUP)

Review	Coverage	Main point used here
Zawacki-Richter et al. [4]	146 articles, AI in higher education	13 first authors from education (8.9%); 2 articles critical on ethics
Wang et al. [3]	Past decade	AI in classroom discourse
Yan et al. [25]	118 papers since 2017	53 LLM use cases; low readiness, weak replicability, privacy
Sharma et al. [26]	55 studies, 2020–2024	LLMs in personalized learning; need for rigorous evaluation
Abdelrahman et al. [27]	Knowledge tracing survey	Early methods to deep learning; research gaps
González-Pérez et al. [28]	36 articles, 2021–2025	Models do not match available data (logs, grades)
Boujmiraz et al. [29]	72 studies, performance prediction	Adoption of explainable AI remains limited

Cohen's $\kappa = 0.69$ against the human standard and ChatGPT 0.33, while both performed unsatisfactorily on fine-grained codes. The authors propose that AI performs the category-level analysis and that educators refine the details.

Discourse codes are not the same as knowledge. Closer to the task of the system, Kortemeyer [20] used GPT-4 without task-specific training to grade 14,186 written student answers to 504 science and electronics items. The 2-way F1 was 0.744 on SciEntsBank and 0.611 on Beetle, which the author describes as comparable to hand-engineered models but worse than pre-trained models with specialized training. The author notes that sending student data to a cloud model raises privacy concerns.

The three studies are difficult to compare: they differ in model generation, task (discourse coding versus answer grading), subject, and language, and none reports how recognition errors would affect the LLM's judgments.

The reviews describe the same promise and the same limits. Wang et al. [3] review AI in classroom discourse over the past decade. Yan et al. [25] reviewed 118 papers on LLM-supported educational tasks and name low technological readiness, weak replicability, and insufficient privacy protection as challenges. Sharma et al. [26] reviewed 55 studies of LLMs in personalized learning and stress the need for rigorous evaluation. For the system, three rules follow: the LLM should rate understanding against explicit, coarse categories; its ratings should be validated against teachers in each subject; and the choice between a cloud and a local model is a trade-off between accuracy and data protection.

A. Adapting an Open Model

The plan of the system provides for a cloud LLM in the first version and for an open LLM fine-tuned on project data in the

second phase. Two observations motivate the second step: a general-purpose model without task-specific training reached only moderate agreement with experts [2], [20], and a local model keeps student data on the server, a concern raised in [20], [25].

Full fine-tuning of a large model is expensive. Hu et al. [11] proposed low-rank adaptation (LoRA), which freezes the pre-trained weights and trains small additional matrices. Compared with full fine-tuning of GPT-3 175B, it reduces the number of trainable parameters 10,000 times and the GPU memory requirement three times. Dettmers et al. [12] combined LoRA with a frozen 4-bit quantized base model (QLoRA). Their method allows a 65-billion-parameter model to be fine-tuned on a single 48 GB GPU while preserving the task performance of 16-bit fine-tuning. These results concern instruction following and not educational assessment, and the model is larger than the 7–9 billion parameter models planned for the system.

Two open datasets of teacher–student dialogue are available for trial runs. MathDial [13] contains about 3,000 one-to-one tutoring dialogues in which human teachers worked with an LLM playing a student who makes typical errors in multi-step mathematical problems. The NCTE Transcripts [14] contain 1,660 grade 4 and 5 mathematics lessons from 317 teachers, with turn-level annotations of discourse moves. Both datasets are in English and in mathematics, and their annotations describe teacher moves and discourse and not the understanding of topics by the student. They can be used to test the training procedure, but they cannot replace examples corrected by teachers.

No study was found that fine-tunes an open LLM to rate students' understanding from lesson transcripts. The gain from fine-tuning and the amount of teacher-corrected data needed are therefore unknown and have to be measured, against the same model without fine-tuning and a cloud model, on lessons not used for training.

V. From Answers to a Knowledge State (RQ3)

A rating of one answer is not yet an estimate of what a student knows. Knowledge tracing updates a learner model from sequences of evidence; the survey of Abdelrahman et al. [27] covers methods from early work to recent deep learning approaches and discusses research gaps. Sarsa et al. [21] evaluated deep knowledge tracing models (DKT, LSTM-DKT, DKVMN, SAKT) against simple baselines, logistic regression, and Bayesian knowledge tracing. The deep models generally performed better, but the margin was often small, and simple baselines could outperform them on accuracy. The results depended on hyperparameters and the choice of metric. Cukurova et al. [22] mined 2,250 minutes of one-to-one online tutoring and separated effective from less effective sessions with an F-measure of 0.89–0.98. The target is the quality of tutoring, but the study shows that sequences of interaction carry information beyond isolated utterances. A review of learning analytics [28] found that models often do not match the available educational data, which is frequently limited to access logs or grades. A review of performance prediction [29] found that the adoption of explainable AI remains limited.

These studies work with structured logs and not with spoken dialogue. The evidence suggests a transparent, rule-based aggregation of dialogue ratings and quiz results into a per-topic index as the first model, with any learned knowledge-tracing model compared against it as a baseline and not assumed to be better.

VI. Trust, Oversight, and Fairness (RQ4)

Teacher-facing AI is adopted only if teachers can check it. Aldino et al. [23] observed 21 higher-education teachers who wrote feedback on a student presentation with a tool that combined a BERT classifier and ChatGPT. Teachers frequently adopted the suggestions of the classifier but extensively revised the ChatGPT drafts, and they raised concerns about tone and trust and said that human editing was needed. In a randomized study with 59 pre-service teachers (30 and 29 in the two groups), Wang et al. [24] found that adding explanations to the AI analysis of classroom dialogue significantly increased trust and technology acceptance, with no significant difference in cognitive load. Both studies are small, measure perceptions over a short period, and involve feedback writing or pre-service teachers and not assessment by practicing teachers.

A broad review points the same way. Zawacki-Richter et al. [4] analyzed 146 articles and found that only 13 of them (8.9%) had a first author from an education department and that only two (1.4%) reflected critically on ethical implications.

Fairness is the other side of trust. The speech studies show that accuracy has to be reported per task and per group [15], [16], that such checks are feasible [17], and that the language is one of the groups that matter [18]. The rate of hallucinated text also differed between speaker groups [8]. A system that worked well for only one of English, Russian, and mixed speech would treat students unequally.

VII. Discussion

A. Comparison of the Studies

Table III shows which stages of the pipeline each research study addresses. No study combines diarization, ASR, dialogue analysis, and knowledge assessment. The broadest [16] covers four of six components without speaker separation or explanations, and the two studies on Kazakh and Russian ASR [18], [19] do not concern education. The studies measure different things with different metrics: Cohen's and Fleiss's κ, F1, WER, CER, DER, correlations, MAE, and perception scales, applied to objects from single words read by children to lessons of adults. Published figures are therefore orientation points and not targets, and component-level and end-to-end results have to be reported separately.

B. Research Gaps

Six gaps follow from the comparison.

1) *Integration and error propagation.* The components have not been combined into one teacher-facing workflow. Errors of recognition, diarization, and denoising are documented [7], [8], [15]–[17], but none of the reviewed

TABLE III
PIPELINE COMPONENTS ADDRESSED BY THE RESEARCH STUDIES

Study	ASR	DIA	DLG	KNW	EXP	FAI
Petscher [15]	✓			✓		✓
Bailey [16]	✓		✓	✓		✓
Wang (diar.) [17]	✓	✓				✓
Makhataeva [18]	✓					✓
Mussakhojayeva [19]	✓					
Long [1]			✓			
Xu [2]			✓			
Kortemeyer [20]			✓	✓		
Sarsa [21]				✓		
Cukurova [22]			✓			
Aldino [23]			✓		✓	
Wang (XAI) [24]			✓		✓	
Total	5	1	7	4	2	4

ASR: speech recognition or verification; DIA: diarization; DLG: dialogue or text analysis; KNW: assessment of knowledge or learning state; EXP: explainability or human oversight; FAI: fairness, validity, or per-group (e.g., per-language) analysis. A mark means a central focus of the study (authors' classification).

LLM studies reports how such errors change an LLM's assessment.

2) *Validation against knowledge.* LLM studies validate against discourse categories [1], [2] or written benchmark answers [20], and not against teachers' judgments of what a student understood in a spoken lesson.
3) *Explainability.* Explanations raise trust in AI labels of dialogue [24], but how to explain an assessment of a student's understanding has not been studied in the reviewed set.
4) *Knowledge over time.* Knowledge tracing is built on structured logs [21], [27], and linking it to conversational evidence remains open.
5) *Language.* The educational evidence comes mainly from English and Chinese settings [1], [2], [15], [16]. The only Kazakh and Russian evaluations [18], [19] use read or crowd-sourced monolingual speech. No evaluation of spontaneous English, Russian, or mixed English–Russian teacher–student dialogue was found, and the examined code-switching work on Whisper concerns another language pair [10].
6) *Adaptation.* No study was found that fine-tunes an open LLM to rate students' understanding from lesson transcripts. The available open dialogue datasets [13], [14] are in mathematics and annotated for teacher moves.

C. Implications for the Proposed System

Fig. 1 maps the reviewed evidence onto the pipeline of the system. The following requirements follow from it.

- *Languages.* The first version covers English, Russian, and mixed lessons. Kazakh is recognized clearly worse [18], [19] and has no evaluation on lessons, so it is deferred; a control WER measurement on a few Kazakh recordings documents the decision.
- *Speech.* Accuracy is measured separately for each language and for mixed speech on the project's own recordings [10], [17], [18]. Denoising is judged by WER and can be switched off [7], pauses are handled explicitly [8], and the teacher can correct speaker labels.
- *LLM analysis.* The LLM rates understanding against explicit categories and cites the time-coded fragments behind each rating [2], [24]. Only anonymized text goes to a cloud model [20], [25].
- *Index.* The index is computed by transparent rules and not by the LLM; learned models serve as a baseline [21], [27].
- *Adaptation.* The training procedure is tested on open dialogues [13], [14] and then applied to the project's own ratings corrected by teachers, with training and control sets separated by lesson [11], [12].

Component metrics (WER, DER, agreement of LLM ratings with teachers, base versus fine-tuned versus cloud model) are reported separately from end-to-end agreement between the system's topic status and teachers' judgments on a control set of lessons. The system supports the teacher's judgment and does not replace it [4], [23].

D. Limitations of This Review

The review is purposive, so other search terms might have retrieved other studies, and the later targeted rounds may favor sources that confirm the gaps found earlier. The core group is limited to journal articles, so it lags behind the fastest-moving ASR and LLM research; the technical group compensates for this only partly, and two of its sources are preprints. Appraisal was qualitative and done by a single rater. Results for specific models (GPT-4, GPT-4o, ChatGPT, Whisper large-v3-turbo) may not hold for newer ones, and the Kazakh and Russian evidence concerns read speech outside the classroom. The figures are quoted from the original articles and were not reproduced.

VIII. CONCLUSION

The reviewed literature supports the feasibility of each stage of the proposed system but not yet of the system as a whole. Speech technologies turn lessons into analyzable text, but accuracy depends on the language and the recording conditions; Kazakh is recognized far less accurately than Russian and English, which is the basis for the scope of the first version (RQ1). LLMs judge dialogue and short answers with moderate accuracy, agreement with experts drops for fine-grained judgments, and parameter-efficient fine-tuning makes adapting an open model feasible, but its effect on rating understanding has not been studied (RQ2). Complex knowledge tracing models do not reliably beat simple baselines (RQ3). Explanations and human oversight are decisive for teacher acceptance, and accuracy has to be checked per group and per language (RQ4). Six gaps remain: integration and error propagation, validation against teachers' judgments, explainability, knowledge over time, evidence for English, Russian, and mixed lessons, and adaptation of open LLMs to rating understanding. The proposed system addresses them by combining speaker-aware recognition of English, Russian, and mixed speech, evidence-based LLM ratings, a transparent knowledge index, and a fine-tuning experiment, and by evaluating each stage against teachers' judgments.

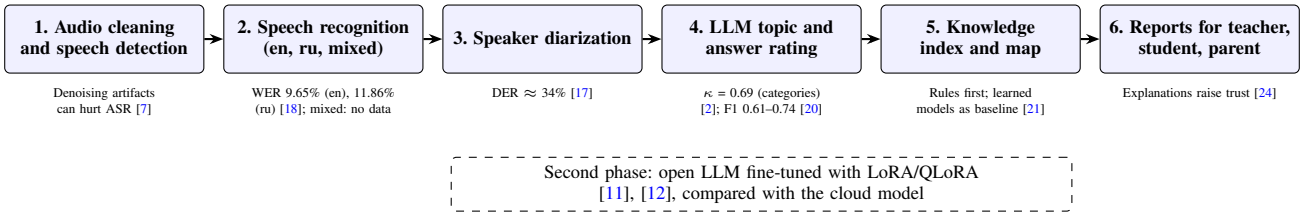


Fig. 1. Pipeline of the proposed system and the literature evidence relevant to each stage.

REFERENCES

[1] Y. Long, H. Luo, and Y. Zhang, "Evaluating large language models in analysing classroom dialogue," *npj Science of Learning*, vol. 9, no. 1, 2024, art. no. 60. [Online]. Available: https://doi.org/10.1038/s41539-024-00273-3

[2] S. Xu, X. Huang, C. K. Lo, G. Chen, and M. S.-Y. Jong, "Evaluating the performance of ChatGPT and GPT-4o in coding classroom discourse data: A study of synchronous online mathematics instruction," *Computers and Education: Artificial Intelligence*, vol. 7, 2024, art. no. 100325. [Online]. Available: https://doi.org/10.1016/j.caeai.2024.100325

[3] D. Wang, Y. Tao, and G. Chen, "Artificial intelligence in classroom discourse: A systematic review of the past decade," *International Journal of Educational Research*, vol. 123, 2024, art. no. 102275. [Online]. Available: https://doi.org/10.1016/j.ijer.2023.102275

[4] O. Zawacki-Richter, V. I. Marín, M. Bond, and F. Gouverneur, "Systematic review of research on artificial intelligence applications in higher education – where are the educators?" *International Journal of Educational Technology in Higher Education*, vol. 16, 2019, art. no. 39. [Online]. Available: https://doi.org/10.1186/s41239-019-0171-0

[5] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. 40th Int. Conf. Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 202, 2023, pp. 28 492–28 518. [Online]. Available: https://proceedings.mlr.press/v202/radford23a.html

[6] H. Bredin, R. Yin, J. M. Coria, G. Gelly, P. Korshunov, M. Lavechin, D. Fustes, H. Titeux, W. Bouaziz, and M.-P. Gill, "pyannote.audio: neural building blocks for speaker diarization," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2020, arXiv:1911.01255. [Online]. Available: https://arxiv.org/abs/1911.01255

[7] K. Iwamoto, T. Ochiai, M. Delcroix, R. Ikeshita, H. Sato, S. Araki, and S. Katagiri, "How bad are artifacts? Analyzing the impact of speech enhancement errors on ASR," in *Proc. Interspeech*, 2022, arXiv:2201.06685. [Online]. Available: https://www.isca-archive.org/interspeech_2022/iwamoto22_interspeech.html

[8] A. Koenecke, A. S. G. Choi, K. X. Mei, H. Schellmann, and M. Sloane, "Careless Whisper: Speech-to-text hallucination harms," in *Proc. ACM Conf. Fairness, Accountability, and Transparency (FAccT)*, 2024, arXiv:2402.08021. [Online]. Available: https://doi.org/10.1145/3630106.3658996

[9] M. T. Agro, A. Kulkarni, K. Kadaoui, Z. Talat, and H. Aldarmaki, "Code-switching in end-to-end automatic speech recognition: A systematic literature review," arXiv preprint arXiv:2507.07741, 2025. [Online]. Available: https://arxiv.org/abs/2507.07741

[10] J. Zhao, H. Shi, C. Cui, T. Wang, H. Liu, Z. Ni, L. Ye, and L. Wang, "Adapting Whisper for code-switching through encoding refining and language-aware decoding," arXiv preprint arXiv:2412.16507, 2024. [Online]. Available: https://arxiv.org/abs/2412.16507

[11] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022, arXiv:2106.09685. [Online]. Available: https://arxiv.org/abs/2106.09685

[12] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, arXiv:2305.14314. [Online]. Available: https://arxiv.org/abs/2305.14314

[13] J. Macina, N. Daheim, S. Chowdhury, T. Sinha, M. Kapur, I. Gurevych, and M. Sachan, "MathDial: A dialogue tutoring dataset with rich pedagogical properties grounded in math reasoning problems," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023, pp. 5602–5621. [Online]. Available: https://aclanthology.org/2023.findings-emnlp.372/

[14] D. Demszky and H. Hill, "The NCTE transcripts: A dataset of elementary math classroom transcripts," in *Proc. 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA)*, 2023, pp. 528–538. [Online]. Available: https://aclanthology.org/2023.bea-1.44

[15] Y. Petscher, J. O'Sullivan, H. W. Catts, A. Edwards, and L. Fitton, "Evaluation of the consistency of a speech verification system with human raters in early literacy screening assessments," *Frontiers in Education*, vol. 11, 2026, art. no. 1671946. [Online]. Available: https://doi.org/10.3389/feduc.2026.1671946

[16] A. L. Bailey, A. Johnson, N. B. Shankar, H. Veeramani, J. A. Washington, and A. Alwan, "Addressing bias in spoken language systems used in the development and implementation of automated child language-based assessment," *Journal of Educational Measurement*, vol. 63, no. 1, 2026, art. no. e12435. [Online]. Available: https://doi.org/10.1111/jedm.12435

[17] J. Wang, S. Dudy, X. He, Z. Wang, R. Southwell, and J. Whitehill, "Optimizing speaker diarization for the classroom: Applications in timing student speech and distinguishing teachers from children," *Journal of Educational Data Mining*, vol. 17, no. 1, pp. 98–125, 2025. [Online]. Available: https://doi.org/10.5281/zenodo.14871875

[18] Z. Makhataeva, N. Atymtay, R. Meiramov, G. Nauryzbaikyzy, K. Sadirova, and H. A. Varol, "Multilingual automatic speech recognition interface for typing: Usability study and performance evaluation for Kazakh, Russian, and English," *Scientific Journal of Astana IT University*, vol. 24, pp. 49–63, 2025. [Online]. Available: https://doi.org/10.37943/24AHNP6638

[19] S. Mussakhojayeva, K. Dauletbek, R. Yeshpanov, and H. A. Varol, "Multilingual speech recognition for Turkic languages," *Information*, vol. 14, no. 2, 2023, art. no. 74. [Online]. Available: https://doi.org/10.3390/info14020074

[20] G. Kortemeyer, "Performance of the pre-trained large language model GPT-4 on automated short answer grading," *Discover Artificial Intelligence*, vol. 4, 2024, art. no. 47. [Online]. Available: https://doi.org/10.1007/s44163-024-00147-y

[21] S. Sarsa, J. Leinonen, and A. Hellas, "Empirical evaluation of deep learning models for knowledge tracing: Of hyperparameters and metrics on performance and replicability," *Journal of Educational Data Mining*, vol. 14, no. 2, pp. 32–102, 2022. [Online]. Available: https://doi.org/10.5281/zenodo.7086179

[22] M. Cukurova, M. Khan-Galaria, E. Millán, and R. Luckin, "A learning analytics approach to monitoring the quality of online one-to-one tutoring," *Journal of Learning Analytics*, vol. 9, no. 2, pp. 105–120, 2022. [Online]. Available: https://doi.org/10.18608/jla.2022.7411

[23] A. A. Aldino, B. Maheshi, Y. Li, Y. Zhou, Y.-S. Tsai, D. Gašević, and G. Chen, "Enhancing learner-centered feedback with AI: teachers' practices and perceptions," *Assessment & Evaluation in Higher Education*, vol. 51, no. 5, pp. 892–910, 2026. [Online]. Available: https://doi.org/10.1080/02602938.2026.2638920

[24] D. Wang, C. Bian, and G. Chen, "Using explainable AI to unravel classroom dialogue analysis: Effects of explanations on teachers' trust, technology acceptance and cognitive load," *British Journal of Educational Technology*, vol. 55, no. 6, pp. 2530–2556, 2024. [Online]. Available: https://doi.org/10.1111/bjet.13466

[25] L. Yan, L. Sha, L. Zhao, Y. Li, R. Martinez-Maldonado, G. Chen, X. Li, Y. Jin, and D. Gašević, "Practical and ethical challenges of large language models in education: A systematic scoping review," *British Journal of Educational Technology*, vol. 55, no. 1, pp. 90–112, 2024. [Online]. Available: https://doi.org/10.1111/bjet.13370

[26] S. Sharma, P. Mittal, M. Kumar, and V. Bhardwaj, "The role of large language models in personalized learning: A systematic review of educational impact," *Discover Sustainability*, vol. 6, no. 1, 2025, art. no. 243. [Online]. Available: https://doi.org/10.1007/s43621-025-01094-z

[27] G. Abdelrahman, Q. Wang, and B. Nunes, "Knowledge tracing: A survey," *ACM Computing Surveys*, vol. 55, no. 11, 2023, art. no. 224. [Online]. Available: https://doi.org/10.1145/3569576

[28] L. I. González-Pérez, F. J. García-Peñalvo, and A. J. Argüelles-Cruz, "Data-driven learning analytics and artificial intelligence in higher education: A systematic review," *IEEE Revista Iberoamericana de Tecnologias del Aprendizaje*, vol. 20, pp. 440–451, 2025. [Online]. Available: https://doi.org/10.1109/RITA.2025.3615512

[29] S. Boujmiraz, H. Darhmaoui, and A. Drissi el maliani, "Predicting student performance: A comprehensive review of machine learning, deep learning, and explainable AI approaches," *Computers and Education: Artificial Intelligence*, vol. 10, 2026, art. no. 100548. [Online]. Available: https://doi.org/10.1016/j.caeai.2026.100548